Changming Liu

Software Engineer, Google, Sunnyvale, CA

Contact Information

Google Scholar: https://scholar.google.com/citations?user=DBc2YBoAAAAJ **Phone:** +1-617-637-6223
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Summary: I did research in automated testing/bug-finding for Linux and embedded systems, specializing in fuzz testing and symbolic execution for low-level system software. Additionally, my industry working experience focuses on building low-level management software for cloud products (e.g., accelerators) for features like confidential computing and RAS (i.e., reliability availability serviceability).

Education

PhD of Cybersecurity, advised by Prof Engin Kirda and Prof Long Lu 2019/09 - 2024/10
Khoury College of Computer Science, Northeastern University Boston, MA, USA

M.Eng of Computer Architecture, advised by Prof Hai Jin 2016/09 - 2019/06
School of Computer Sci & Tech, Huazhong University of Sci & Tech Wuhan, China

B.Eng of Information Security, advised by Prof Deqing Zou 2012/09 - 2016/06
School of Computer Sci & Tech, Huazhong University of Sci & Tech Wuhan, China

Experience

Software Engineer 2024/12 - Now
Accelerator Security, Google Cloud Sunnyvale, CA, USA
- Integrating confidential computing into cloud accelerator products.
- Facilitating faster integration of experimental accelerators.

Research Intern 2021/05 - 2021/08
T.J. Watson Research Center, IBM Information Security Research Yorktown Heights, NY, USA
- Building infrastructure for cloud services to automatically infer the microservices' identities and establish trust.

Research Intern 2017/09 - 2018/03, 2018/09 - 2019/03
Software analytics group, Microsoft Research Asia Beijing, China
- Designing and implementing fuzz testing algorithm to detect concurrency bugs.
- Integrating deep-learning-based object detection framework into outlook's anti-phishing system.

Research Assistant 2015/07 - 2015/09
Computer Science Department, University of Hong Kong Hong Kong SAR, China
- Research paper reading and implementation about privacy-preserving genome applications.

Publications

- Changming Liu, Alejandro Mera, Engin Kirda, Meng Xu. **DRIFT: Debug-based Trace Inference for Firmware Testing**. In proceedings of the 40th IEEE/ACM International Conference on Automated Software Engineering (ASE 2025).
- Changming Liu, Alejandro Mera, Engin Kirda, Meng Xu, Long Lu. **CO3: Concolic Co-execution for Firmware**. In proceedings of the 33rd USENIX Security Symposium (USENIX Security 24).

- Alejandro Mera, Changming Liu, Ruimin Sun, Engin Kirda, Long Lu **SHiFT: Semi-hosted Fuzz Testing for Embedded Applications**. In proceedings of the 33rd USENIX Security Symposium (USENIX Security 24).
- Bo Feng, Meng Luo, Changming Liu, Long Lu and Engin Kirda, **AIM: Automatic Interrupt Modeling for Dynamic Firmware Analysis** in IEEE Transactions on Dependable and Secure Computing, doi: 10.1109/TDSC.2023.3339569.
- Zhichuang Sun, Runmin Sun, Changming Liu, Chowdhury Amrita Roy, Somesh Jha and Long Lu. 2023. **ShadowNet: A Secure and Efficient System for On-device Model Inference** in 2023 IEEE Symposium on Security and Privacy (SP), San Francisco, CA, USA, 2023 pp. 1596-1612.
- Changming Liu, Xiaojing Ma, Sixing Cao, Jiayun Fu, and Bin B. Zhu. 2022. **Privacy-preserving Motion Detection for HEVC-compressed Surveillance Video**. ACM Trans. Multimedia Comput. Commun. Appl. 18, 1, Article 23 (January 2022), 27 pages.
- Changming Liu, Yaohui Chen, and Long Lu. 2021. **KUBO: Precise and Scalable Detection of User-triggerable Undefined Behavior Bugs in OS Kernel**. In Proceedings of the 28th Network and Distributed System Security Symposium(NDSS 2021), Virtual Conference, 21st February - 25th February 2021.
- Peng Luo, Deqing Zou, Yajuan Du, Hai Jin, Changming Liu, and Jinan Shen. 2020. **Static detection of real-world buffer overflow induced by loop**. Computers & Security, Volume 89, 2020, 101616, ISSN 0167-4048.
- Changming Liu, Deqing Zou, Peng Luo, Bin B. Zhu, and Hai Jin. 2018. **A Heuristic Framework to Detect Concurrency Vulnerabilities**. In Proceedings of the 34th Annual Computer Security Applications Conference (ACSAC '18). Association for Computing Machinery, New York, NY, USA, 529–541
- Xiaojing Ma, Changming Liu, Sixing Cao, and Bin B. Zhu. 2018. **JPEG Decompression in the Homomorphic Encryption Domain**. In Proceedings of the 26th ACM international conference on Multimedia (MM '18). Association for Computing Machinery, New York, NY, USA, 905–913.
- Liu Changming, Fu Cai, Xu Deliang, Sun Lin, Han Lansheng. 2015. **An Energy-Balanced WSN Algorithm Based on Active Hibernation and Data Recovery**. In Proceedings of Algorithms and Architectures for Parallel Processing. ICA3PP 2015. Lecture Notes in Computer Science, vol 9528. Springer, Cham.

Honors & Awards

Excellent Intern Award(Top 20%)	Microsoft Research Asia, 2019
National Scholarship	Ministry of Education of China, 2018
Excellent Intern Award(Top 20%)	Microsoft Research Asia, 2018
Graduate with honor	Huazhong University of Sci & Tech, 2016
Merit Student	Huazhong University of Sci & Tech, 2015
National Scholarship	Ministry of Education of China, 2015

Misc

Reviewership:	Transactions on Dependable and Secure Computing; Computer & Security; Journal of Systems Architecture; NDSS AE 2027;Usenix Sec AE 2026; NDSS AE 2026; Usenix Sec AE 2025; NDSS 2021 (External); CCS 2021 (External)
Programming:	C/C++;Python;
Hobbies:	Badminton and violin.
Bug Hunt:	Find 14 new undefined behaviors bugs in the Linux kernel that are either acknowledged or patched.